

PHYSICS CLASS 9

Complete Subjective Solutions

FBISE SSC-I | 1st Annual 2026 | Paper D

Inclusive Scheme of Studies 2024

Sections B and C - all questions and all OR alternatives

Prepared for: Fakhar STEM Sphere

Coverage: Q2(i-x), both alternatives; Q3-Q5, both alternatives.

Paper marks: Section B: 30; Section C: 18; total: 48.

These answers use simple English, the essential scientific terms, complete calculations and labelled diagrams. They address the marks shown on the supplied paper. Actual marks depend on the examiner and the answer written.

Numerical convention: Use $g = 10 \text{ m/s}^2$, as printed on the paper. Show given data, formula, substitution and the answer with its unit.

Revision: Learn the definitions and bold results, practise the working, and redraw the graph and hydraulic-lift sketch. Both OR alternatives are solved for preparation; follow the choice instructions in the actual examination.

SECTION B | Q2(i) and Q2(ii)

Q2(i) - Briefly describe astronomy, particle physics and geophysics. [1 × 3]

(a) Astronomy: The study of celestial bodies, such as stars, planets and galaxies, and phenomena in the universe.

(b) Particle physics: The study of the fundamental particles of matter and the forces or interactions between them.

(c) Geophysics: The study of Earth using the principles of physics, including its interior, gravity, magnetic field and earthquakes.

OR - Determine total distance, total time and average speed from the car diagram. [1 × 3]

Reading the diagram: The car passes positions 0 m, 6 m, 14 m and 24 m. The three successive journey intervals are 2 s, 4 s and 2 s.

(a) Total distance:

$$d = (6 - 0) + (14 - 6) + (24 - 14) = 6 + 8 + 10 = \mathbf{24 \text{ m.}}$$

(b) Total time:

$$t = 2 + 4 + 2 = \mathbf{8 \text{ s.}}$$

(c) Average speed:

$$\text{Average speed} = \text{total distance} / \text{total time} = 24/8 = \mathbf{3 \text{ m/s.}}$$

Exam tip: The position labels are cumulative distances; do not add $6 + 14 + 24$. Use total distance divided by total time, rather than the simple average of the three segment speeds.

Q2(ii) - How does an electromagnet improve the working of an electric bell? [3]

1. When the switch is pressed, current flows through a coil wound around a soft-iron core. It becomes an electromagnet and attracts the iron armature, causing the attached hammer to strike the gong.
2. The armature's movement breaks the electrical contact. Current stops, the electromagnet loses most of its magnetism, and a spring pulls the armature back.
3. The contact is made again and the process repeats rapidly, producing continuous ringing. The electromagnet can be switched on and off automatically, which enables these repeated strikes.

OR - How does a rapid rise or fall in the barometer column help predict weather? [3]

The height of a barometer's liquid column indicates atmospheric pressure because $P = \rho gh$.

Rapid rise: Indicates increasing pressure and often the approach of a high-pressure system, so clearing or drier weather is likely.

Rapid fall: Indicates decreasing pressure and often the approach of a low-pressure system, so cloud, rain or stormy weather is likely.

The rate of change helps indicate how quickly the weather system is changing; these are general weather tendencies.

SECTION B | Q2(iii) and Q2(iv)

Q2(iii) - A digital balance gives slightly different readings for the same stone. Identify the error and explain how to fix it. [1 + 2]

(a) Error: This is a **random error**, because repeated readings fluctuate unpredictably even though the stone's mass is unchanged.

(b) Reduction: Place the balance on a stable, level surface, avoid air currents and vibrations, and wait for each reading to settle. Take several readings and calculate their average: **mean mass = sum of readings / number of readings**. Averaging reduces the influence of random variation; it does not guarantee complete removal of random error.

OR - Explain how liquid rising in a straw or dropper shows that the atmosphere exerts pressure. [3]

Straw: Sucking removes some air and lowers the pressure inside the straw. Atmospheric pressure acting on the exposed liquid surface is then greater than the pressure inside the straw. This pressure difference pushes the liquid up the straw.

Dropper: Squeezing its bulb expels air. When the tip is immersed and the bulb is released, the bulb expands and the pressure inside falls. Atmospheric pressure on the liquid surface pushes liquid into the dropper.

In both cases, liquid rises because of an external atmospheric pressure acting against a lower internal pressure.

Q2(iv) - Define efficiency and give the formula for percentage efficiency. [2 + 1]

Efficiency is the ratio of useful output energy to total input energy. It measures how much of the supplied energy is converted into the desired useful form.

Percentage efficiency = (useful output energy / total input energy) × 100%.

For rates of energy transfer, the equivalent expression is (useful output power / total input power) × 100%. Efficiency is a ratio and has no physical unit.

OR - Define velocity. Explain why speed is scalar and velocity is vector. [1 + 2]

Velocity is the rate of change of displacement: **average velocity = displacement / time.**

Speed is scalar: It specifies only how fast a body moves, with no direction; for example, 10 m/s.

Velocity is vector: It specifies both magnitude and direction; for example, 10 m/s east. A change in direction changes velocity even when speed stays constant.

SECTION B | Q2(v) and Q2(vi)

Q2(v) - Name any three fundamental forces and briefly describe each. [1 × 3]

1. Gravitational force: The attractive force between bodies with mass, such as Earth's attraction on objects.

2. Electromagnetic force: The force associated with electric charges and magnetic fields; it may cause attraction or repulsion.

3. Strong nuclear force: The fundamental interaction that binds quarks; its residual effect binds protons and neutrons together in atomic nuclei.

Another acceptable choice: The **weak nuclear force** is responsible for processes such as beta decay. Only three forces are requested.

OR - Differentiate between precision and accuracy. [3]

Precision	Accuracy
The closeness of repeated measurements to one another.	The closeness of a measurement to the accepted or true value.
High precision means a small spread in repeated readings.	High accuracy means a small difference from the accepted value.
Readings of 12.1, 12.1 and 12.2 g are precise, but inaccurate if the true mass is 10.0 g.	A measured mass of 10.0 g is accurate if the true mass is 10.0 g.

Q2(vi) - What is average orbital speed? Why does a larger orbital radius increase the distance in one orbit? [1 + 2]

Average orbital speed is the total distance travelled along an orbit divided by the time taken to complete it.

For a circular orbit of radius r and period T , $v = 2\pi r/T$.

The distance travelled in one complete circular orbit is its circumference, $d = 2\pi r$. Therefore, distance is directly proportional to radius: a larger radius gives a larger circumference. If the radius doubles, the distance in one orbit also doubles.

OR - Define internal energy. Why does heating increase it? [1 + 2]

Internal energy is the total microscopic kinetic and potential energy of the particles of a substance.

When energy is supplied by heating, it can increase the average kinetic energy of particles, so they move or vibrate more vigorously and the temperature rises. During a change of state, the supplied energy can increase their potential energy by changing their separation and overcoming intermolecular attractions, even though temperature remains constant.

Concept clarification: This is the expected explanation when the supplied heat is retained as internal energy. The word "always" is not universally true: in general, $\Delta U = Q - W$, where W is work done by the substance. If all the supplied heat becomes work, internal energy need not increase.

SECTION B | Q2(vii) and Q2(viii)

Q2(vii) - Why do high-pressure areas generally have clear skies and low-pressure areas often have rain? [1.5 + 1.5]

High pressure: Air generally sinks. As it descends, it is compressed and warms, so its relative humidity decreases. This discourages condensation and cloud formation, producing generally clear skies.

Low pressure: Air generally rises. It expands and cools as pressure decreases with height. If it cools to its dew point, water vapour condenses into droplets, producing clouds and possibly rain.

OR - State Newton's third law and give three daily-life examples. [1.5 + 1.5]

Law: For every action, there is an equal and opposite reaction. When one body exerts a force on another, the second exerts an equal force in the opposite direction on the first. These forces act on **different bodies**.

1. Walking: The foot pushes the ground backwards; the ground pushes the person forwards.

2. Swimming: The swimmer pushes water backwards; the water pushes the swimmer forwards.

3. Jumping: The person pushes the ground downwards; the ground pushes the person upwards.

Q2(viii) - Compare paramagnetic and diamagnetic materials. [1 × 3]

Paramagnetic materials	Diamagnetic materials
Are weakly attracted by an external magnetic field.	Are weakly repelled by an external magnetic field.
Their induced magnetisation is in the direction of the applied field.	Their induced magnetisation is opposite to the applied field.
Examples: aluminium and platinum.	Examples: copper and bismuth.

OR - Define base and derived quantities and give an example of each. [1.5 + 1.5]

Base quantities are independent physical quantities chosen as the basis of a system of measurement. Example: **length**, whose SI base unit is the metre (m).

Derived quantities are expressed as combinations of base quantities. Example: **speed = distance/time**, whose SI unit is m/s, derived from length and time.

SECTION B | Q2(ix) and Q2(x)

Q2(ix) - Find the impulse on a 2 kg object whose velocity changes from 3 m/s to 7 m/s. [3]

Given: $m = 2 \text{ kg}$; $u = 3 \text{ m/s}$; $v = 7 \text{ m/s}$, in the same direction.

Required: Impulse J .

Formula: $J = \Delta p = m(v - u)$.

Substitution: $J = 2(7 - 3) = 2 \times 4$.

Answer: $J = 8 \text{ N s}$, equivalent to **8 kg m/s**, in the direction of the increase in velocity.

OR - Why are soft magnetic materials preferred for magnetic shielding? [3]

- 1. High permeability:** Soft magnetic materials, such as soft iron or suitable nickel-iron alloys, provide an easy path for magnetic flux.
- 2. Shielding action:** A shield made from them guides much of the external magnetic flux through its material and around the protected space, reducing the magnetic field inside.
- 3. Low retentivity and coercivity:** They magnetise and demagnetise easily and retain little unwanted magnetism when the external field is removed. This helps protect sensitive laboratory instruments and electrical equipment from magnetic interference.

Q2(x) - Give three reasons why mercury is preferred as a thermometric material. [1 × 3]

- 1.** It expands fairly uniformly with temperature over its useful range, allowing reliable calibration.
- 2.** It does not wet glass, so it forms a clear meniscus and does not stick to the capillary walls.
- 3.** It is shiny and opaque, making its level easy to see.

These are the requested physical properties of mercury in a traditional liquid-in-glass thermometer.

OR - Define pressure. Why does a sharp knife cut better than a blunt knife? [1 + 2]

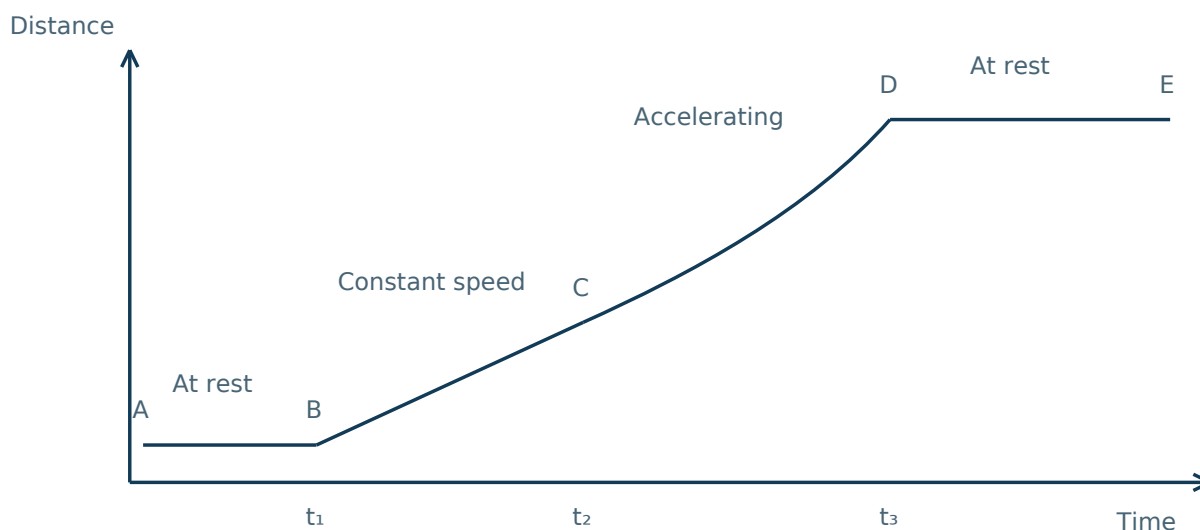
Pressure is the normal force acting per unit area: $P = F/A$. Its SI unit is pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$.

A sharp knife has a smaller contact area at its cutting edge than a blunt knife. For the same applied force, a smaller area gives greater pressure. Thus, the sharp edge exerts greater pressure on the material and cuts it more easily.

SECTION C | Q3 - Distance-time graph [4 + 2]

A car starts from rest, moves with constant speed, then accelerates and finally comes to rest. Draw its distance-time graph and explain the gradients.

(a) Graph: Time is shown on the horizontal axis and cumulative distance on the vertical axis. An initial horizontal portion represents the initial state of rest; the required stages then follow.



AB - Initial rest: Distance remains constant, so the graph is horizontal.

BC - Constant speed: Distance increases uniformly, so the graph is a straight rising line.

CD - Acceleration: Distance increases more rapidly with time, so the curve becomes progressively steeper.

DE - Final rest: Distance remains constant again, so the graph is horizontal.

(b) Meaning of gradient:

Speed = gradient of the distance-time graph = $\Delta \text{distance} / \Delta \text{time}$.

For AB and DE, gradient = 0, so speed = 0.

For BC, gradient is positive and constant, so speed is constant.

For CD, gradient increases with time, so speed increases.

Graph interpretation: The question gives no numerical times or distances, so the graph is schematic. Corners represent idealised changes between stages. If gradual braking before final rest is intended, insert a short rising curve whose slope decreases to zero before DE. Distance must not fall when the car stops; only the slope falls to zero. Do not draw a velocity-time graph in place of the requested distance-time graph.

SECTION C | Q3 - OR: Football numerical [3 + 3]

A 450 g football is kicked vertically upwards at 22 m/s. Find its maximum height without air resistance, and the work against resistance if it rises only 20 m.

Given: $m = 450 \text{ g} = 0.450 \text{ kg}$; $u = 22 \text{ m/s}$ upwards; $g = 10 \text{ m/s}^2$. At maximum height, $v = 0$.

(a) Maximum height without air resistance:

Take upwards as positive, so $a = -g = -10 \text{ m/s}^2$.

$$v^2 = u^2 + 2ah.$$

$$0 = 22^2 - 2 \times 10 \times h.$$

$$h = 22^2 / (2 \times 10) = 484 / 20.$$

Maximum height = 24.2 m.

(b) Work done against air resistance:

At the actual highest point, $h = 20 \text{ m}$ and the ball's kinetic energy is zero. Initial kinetic energy is converted into gravitational potential energy and energy dissipated against air resistance.

Initial kinetic energy:

$$\text{K.E.} = \frac{1}{2}mu^2 = \frac{1}{2} \times 0.450 \times 22^2 = \mathbf{108.9 \text{ J}}.$$

Gain in gravitational potential energy:

$$\text{P.E.} = mgh = 0.450 \times 10 \times 20 = \mathbf{90 \text{ J}}.$$

Work against resistance:

$$W = \text{initial K.E.} - \text{gain in P.E.}$$

$$W = 108.9 - 90 = \mathbf{18.9 \text{ J}}.$$

Final answers: (a) **24.2 m**; (b) **18.9 J**.

Sign note: The work done **against** resistance is positive 18.9 J; the work done **by** air resistance on the ball is -18.9 J.

SECTION C | Q4 - Friction [2 + 2 + 2]

(a) Define friction. (b) Explain energy dissipation in machine parts. (c) Explain two methods to reduce friction.

(a) Friction:

Friction is a contact force that opposes relative motion, or the tendency of relative motion, between surfaces in contact. It acts along the surfaces. For example, friction between a sliding block and a table opposes the sliding motion.

(b) Energy dissipation as heat:

When machine parts rub against each other, friction opposes their relative movement. Work must be done against this force. The rubbing and microscopic deformation transfer mechanical energy into the internal energy of the parts and their surroundings, increasing their temperature. Therefore, less of the input energy is available as useful mechanical output, and machine efficiency decreases.

For constant friction f over a sliding distance d , **energy dissipated = fd** . Energy is conserved; it is converted into a less useful form.

(c) Two methods:

1. Lubrication: Put oil or grease between moving parts. The lubricant forms a layer that reduces direct contact and interlocking of surface irregularities, reducing friction and wear.

2. Ball bearings: Use balls or rollers between moving parts, such as a shaft and its support. This replaces much of the sliding friction with rolling friction, which is generally smaller.

SECTION C | Q4 - OR: Permanent magnets [3 + 3]

Discuss any two applications of permanent magnets in detail.

Application 1: Magnetic compass

Construction: A compass contains a light permanent magnetic needle mounted on a low-friction pivot. It is free to rotate in a horizontal plane.

Working: Earth's magnetic field exerts a turning effect on the needle. It aligns approximately with the horizontal component of Earth's field, with its north-seeking end pointing approximately towards geographic north.

Use: The marked direction allows travellers, sailors and surveyors to find north and determine other directions. The needle retains its magnetism and requires no electric current.

Application 2: Loudspeaker

Construction: A loudspeaker has a strong permanent magnet that produces a steady magnetic field in a narrow gap. A light voice coil attached to a cone lies in this gap.

Working: The changing audio current flows through the voice coil. The interaction between the coil's current and the permanent magnetic field produces a force on the coil. As the current changes in magnitude and direction, the force changes, moving the coil and cone backwards and forwards.

Use: The vibrating cone produces pressure variations in the air, creating sound waves that reproduce the audio signal. The permanent magnet provides the steady field without a separate field-coil current.

Exam tip: For each application, identify the permanent magnet, explain its role in the working, and state the useful result. Naming two devices alone is insufficient for six marks.

SECTION C | Q5 - Centripetal force [1 + 1 + 4]

Define centripetal force, give its formula and calculate it for a 10 kg ball moving at 25 m/s in a circle of radius 1.5 m.

Definition: Centripetal force is the resultant force directed towards the centre of a circular path that keeps a body moving along that path.

Mathematical form: $F_c = mv^2/r$, where m is mass, v is speed and r is radius.

Given: $m = 10 \text{ kg}$; $v = 25 \text{ m/s}$; $r = 1.5 \text{ m}$.

Required: Centripetal force F_c .

Formula: $F_c = mv^2/r$.

Substitution: $F_c = (10 \times 25^2)/1.5$.

$F_c = (10 \times 625)/1.5 = 6250/1.5$.

Answer: $F_c = 4166.7 \text{ N} \approx 4.17 \times 10^3 \text{ N}$, directed towards the centre.

Check: $a_c = v^2/r = 625/1.5 = 416.7 \text{ m/s}^2$.

$F_c = ma_c = 10 \times 416.7 \approx 4167 \text{ N}$.

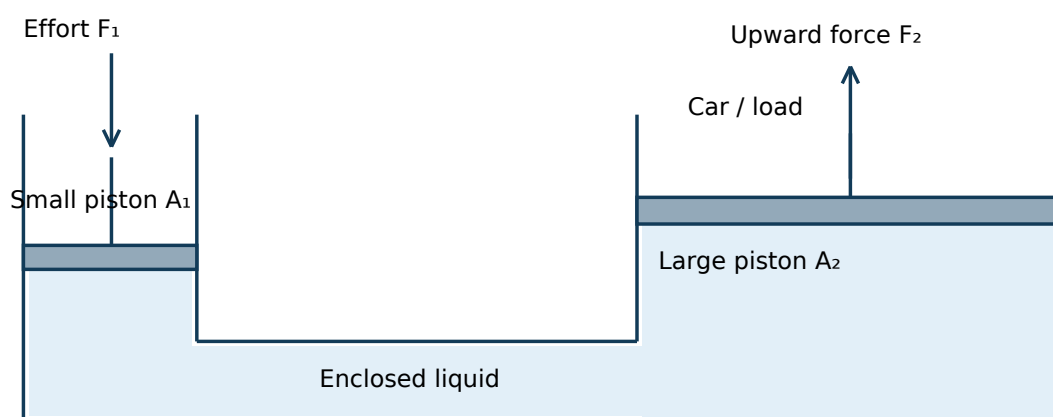
Exam tip: Square the speed, not the mass. The force changes the direction of velocity even if the speed remains constant.

SECTION C | Q5 - OR: Hydraulic car lift [2 + 2 + 1 + 1]

State Pascal's law, explain a hydraulic car lift, derive the force relation and draw a sketch.

Pascal's law: A change in pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the walls of its container.

Construction and working: A hydraulic lift has two connected cylinders filled with liquid. A small piston of area A_1 receives an effort force F_1 , while a large piston of area A_2 supports the car. The force on the small piston increases the pressure in the liquid. This pressure is transmitted to the large piston, producing an upward force F_2 . Since A_2 is larger than A_1 , the output force is greater, allowing a small effort to lift a heavy car.



Mathematical explanation:

Pressure applied at the small piston: $P_1 = F_1/A_1$.

Pressure transmitted to the large piston: $P_2 = F_2/A_2$.

By Pascal's law, for an ideal lift with pistons at the same level, $P_1 = P_2$.

$F_1/A_1 = F_2/A_2$, therefore $F_2 = F_1(A_2/A_1)$.

Thus, $F_2/F_1 = A_2/A_1$. If A_2 is 10 times A_1 , the output force is 10 times the input force. To support a car of mass m , $F_2 = mg$, so $F_1 = mg(A_1/A_2)$.

Energy note: The force gain does not create energy. In an ideal lift, the small piston moves a greater distance: $A_1d_1 = A_2d_2$, so input work equals output work.

Rapid revision | Final numerical answers

Question	Final answer
Q2(i), OR - car	Distance = 24 m; time = 8 s; average speed = 3 m/s
Q2(ix) - impulse	8 N s = 8 kg m/s
Q3, OR (a) - football	Maximum height without resistance = 24.2 m
Q3, OR (b) - football	Work against resistance = 18.9 J
Q5 - centripetal force	4166.7 N $\approx$ 4.17 $\times$ 10 ³ N, towards the centre

Essential formulas

Average speed = total distance / total time

Average velocity = displacement / time

Percentage efficiency = (useful output energy / input energy) $\times$ 100%

Circular orbital speed = $2\pi r/T$

Impulse = $\Delta p = m(v - u)$

Pressure = F/A

Kinetic energy = $\frac{1}{2}mv^2$

Gravitational potential energy gain = mgh

Equation of motion: $v^2 = u^2 + 2as$

Centripetal force = mv^2/r

Hydraulic lift: $F_1/A_1 = F_2/A_2$

Final presentation check: Number the answer correctly; answer every requested subpart; show numerical working and units; give reasons where asked; and draw neat labelled diagrams. In the actual paper, attempt the required OR alternative rather than spending time writing both.